AQA 3.2.2 and 3.2.3 Tests for ions (required practical 4)

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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18 marks · about 25 minutes

Key information

- Silver halides with $NH_3(aq)$: AgCl dissolves in dilute, AgBr only in concentrated, AgI in neither
- OH^-: red litmus turns blue / universal indicator turns blue-purple
- Group 2 cations: NaOH(aq) gives white $Mg(OH)_2$; $H_2SO_4(aq)$ gives white $BaSO_4$ with Ba^{2+}
- NH_4^+: warm with NaOH(aq), damp red litmus turns blue; $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$
- CO_3^{2-}: dilute acid gives effervescence; CO_2 turns limewater cloudy

Section A: Short-answer questions

01 Which observation would confirm that ammonia gas is released when solid ammonium chloride is warmed with solid calcium hydroxide?

[1 mark]

A Damp blue litmus paper turns red when touched onto the solid mixture.
B Damp red litmus paper turns blue when touched onto the solid mixture.
C Damp blue litmus paper turns red when held just above the solid mixture.
D Damp red litmus paper turns blue when held just above the solid mixture.

02 State the observation when dilute hydrochloric acid is added to solid calcium carbonate, and give a test for the gas produced.

[2 marks]

03 Identify a reagent that could be used to indicate the presence of iodide ions in an aqueous solution and describe the observation made.

[2 marks]

04 Give the ionic equation, including state symbols, for the formation of a white precipitate when sodium hydroxide solution is added to a solution containing magnesium ions.

[1 mark]

05 What is the most suitable reagent for detecting the presence of carbonate ions in the presence of an excess of sulfate ions?

[1 mark]

A dilute NaOH(aq)
B dilute $H_2SO_4(aq)$
C $BaCl_2(aq)$
D NaCl(aq)

Section B: Longer questions

06 An aqueous solution Y is known to contain one type of Group 2 metal ion and one type of negative ion.

Aqueous solutions of sulfuric acid and magnesium nitrate are added to separate samples of solution Y. The observations are shown in the table.

Solution added	Observation with solution Y
Sulfuric acid	A white precipitate forms
Magnesium nitrate	A white precipitate forms

06.1 Suggest the identity of the Group 2 metal ion present in solution Y. Write an ionic equation, including state symbols, for the reaction that takes place when sulfuric acid is added to solution Y.

[2 marks]

06.2 Suggest the identity of the negative ion present in solution Y. Write an ionic equation, including state symbols, for the reaction that takes place when magnesium nitrate is added to solution Y.

[2 marks]

07 A student completes a number of test-tube reactions on solutions A, B and C. The table shows the student's observations.

	Test 1 Add $H_2SO_4(aq)$	Test 2 Warm with NaOH(aq)	Test 3 Add acidified $AgNO_3(aq)$
A	white precipitate	no visible change	no visible change
B	effervescence	a gas is formed that turns damp red litmus blue	effervescence
C	no visible change	no visible change	off-white precipitate

07.1 Suggest the identity of the positive ion in solution A. Give the simplest ionic equation for the formation of the white precipitate in Test 1 for solution A.

[2 marks]

07.2 Different gases are formed when solution B reacts in Test 1 and in Test 2. Suggest the identity of each gas. Give the simplest ionic equation for the formation of the gas in Test 2.

[2 marks]

07.3 The student thinks that solution C contains either chloride ions or bromide ions. Describe a further test, or tests, to show whether solution C contains chloride or bromide ions.

[3 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**D**	*Source: 7405/3 Jun 2018 Q20*	1
02	Effervescence / bubbles Gas turns limewater cloudy / milky	*Source: Practice question (written for this set)*	2
03	$AgNO_3$ yellow precipitate	*Source: 7405/1 Jun 2017 Q01.4* Or Cl_2 / Br_2: brown solution / black ppt M2 dependent on correct M1	2
04	$Mg^{2+}(aq) + 2OH^-(aq) \rightarrow Mg(OH)_2(s)$	*Source: Practice question (written for this set)*	1
05	**B**	*Source: 7405/3 Jun 2019 Q19*	1
06.1	Ba^{2+} or Sr^{2+} $SO_4^{2-}(aq) + Ba^{2+}(aq) \rightarrow BaSO_4(s)$	*Source: 7404/1 Jun 2017 Q07*	2
06.2	OH^- $Mg^{2+}(aq) + 2OH^-(aq) \rightarrow Mg(OH)_2(s)$		2
07.1	Ba^{2+} $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$	*Source: 7404/1 Nov 2021 Q04* Accept Ca^{2+} / Sr^{2+} / Pb^{2+} (with consequential equation) Ignore state symbols	2
07.2	Test 1 CO_2 **and** Test 2 NH_3 $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$	Both gases needed for first mark Allow SO_2 for Test 1	2
07.3	M1 add dilute ammonia solution (to the precipitate) M2 if the precipitate dissolves, chloride ions are present M3 if the precipitate does not dissolve, bromide ions are present	Allow M3 if concentrated ammonia then dissolves it Alternative: add chlorine — no visible change for chloride; orange-brown solution for bromide	3